

Project 1 - API Automation Framework (Notes)

Goal: Maintainable REST API tests in Python for an SDET portfolio.

Repo: api-automation-pytest

Target API: JSONPlaceholder (public demo, no auth).

Stack: Python, Pytest, requests, jsonschema, pytest-html, GitHub Actions.

Architecture:

tests call fixtures in conftest, which create an ApiClient.

ApiClient wraps requests.Session with base URL, headers, and timeout.

Default timeout is 10 seconds.

Payloads live in data/payloads. Schemas live in data/schemas.

Coverage:

About 77 automated tests total.

GET and parametrize coverage around 33 cases.

Write operations POST PUT PATCH DELETE around 12 cases.

JSON Schema validation around 12 cases.

Negative and edge cases around 20 cases.

Important honesty: JSONPlaceholder fakes writes. It echoes data and

returns success-like codes but does not really persist new records.

So write tests assert status and response body echo, not GET-after-POST.

Negative testing note:

Unknown resource ids usually return HTTP 404.

A valid filter with zero matches often returns HTTP 200 and an empty list.

Those two outcomes mean different things and should not be mixed up.

CI: GitHub Actions runs pytest on push and uploads an HTML report artifact.